10.25 Spinor Representations of the Lorentz Group

This note connects the proper orthochronous Lorentz group $SO^+(1,3)$, its Lie algebra $\mathfrak{so}(1,3)$, its universal cover $SL(2,\mathbb{C})$, and the classification of its finite-dimensional representations. Conventions and the algebra

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \qquad [J_i, K_j] = i\epsilon_{ijk}K_k, \qquad [K_i, K_j] = -i\epsilon_{ijk}J_k$$

are those of §10.3; a group element near the identity is $\exp(i\theta_i J_i + i\eta_i K_i)$, with rotation angles θ_i and rapidities η_i. With this sign θ_i and η_i are minus the angle and rapidity of LecNote2 p. 3, whose $\delta\omega$ give $U = I - i\,\delta\theta\,\hat{n}\cdot\mathbf{J} + i\,\delta\eta\,\hat{n}\cdot\mathbf{K}$ with Srednicki's $\mathbf{K}$ (so $e^{-i\theta\hat{n}\cdot\mathbf{J}}$ is QM's rotation by θ): here $\exp(i\theta J_3)$ rotates by $-\theta$ about the z axis. The matrix L below is Srednicki's for the same Λ. $SO^+(1,3)$ is the group of real 4×4 matrices Λ that preserve $g = \mathrm{diag}(-1,+1,+1,+1)$ with $\det\Lambda = 1$ and $\Lambda^0{}_0 \geq 1$.

The Template: $SO(3)$ and $SU(2)$

- **Algebras (local).** The Lie algebras are isomorphic, $\mathfrak{su}(2) \cong \mathfrak{so}(3)$: the same three generators J_i with $[J_i, J_j] = i\epsilon_{ijk}J_k$.
- **Groups (global).** The groups are not isomorphic. $SO(3)$ (real orthogonal 3×3 matrices) acts on vectors; $SU(2)$ (complex unitary 2×2 matrices of unit determinant) acts on 2-component spinors. $SU(2)$ is the **universal double cover** of $SO(3)$: the map is 2-to-1, with U and $-U$ giving the same rotation.

The double cover is visible in the action on a vector $\mathbf{x}$ written as the matrix $X = \mathbf{x}\cdot\boldsymbol{\sigma}$:

$$X \to X' = UXU^\dagger, \qquad (-U)X(-U)^\dagger = UXU^\dagger.$$

Physically, a 2π rotation returns a vector to itself but multiplies a spinor by -1; a 4π rotation is needed to return a spinor to itself. The supplement GroupTheoryIntro (section Lie Algebra) constructs both representations of this algebra, the 3×3 generators $(J_i)_{jk} = -i\epsilon_{ijk}$ and the 2×2 ones $J_i = \sigma_i/2$.

The Lorentz Case: $SO^+(1,3)$ and $SL(2,\mathbb{C})$

The same pattern holds with six generators (three rotations J_i, three boosts K_i):

- **Algebras (local).** As 6-dimensional *real* Lie algebras, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})$, where $\mathfrak{sl}(2,\mathbb{C})$ is regarded as a real algebra with basis $\{\sigma_i/2,\ i\sigma_i/2\}$.

- **Groups (global).** $SL(2,\mathbb{C})$, the complex 2×2 matrices of unit determinant, is the universal double cover of $SO^+(1,3)$: M and $-M$ map to the same Λ. Its defining representation and the complex conjugate of it are the 2-component Weyl spinors.

A Common Error: $\mathfrak{so}(1,3) \ncong \mathfrak{su}(2)\oplus\mathfrak{su}(2)$

It is tempting to write the Lorentz algebra as two copies of $\mathfrak{su}(2)$. As real algebras this is wrong: $\mathfrak{su}(2)\oplus\mathfrak{su}(2)\cong\mathfrak{so}(4)$, the algebra of *Euclidean* rotations in four dimensions. The difference is one sign. Take two commuting $\mathfrak{su}(2)$'s with generators A_i, B_i, and set $J_i = A_i + B_i$, $K_i = A_i - B_i$. Then

$$[K_i, K_j] = [A_i, A_j] + [B_i, B_j] = i\epsilon_{ijk}(A_k + B_k) = +i\epsilon_{ijk}J_k,$$

whereas the Lorentz algebra has $-i\epsilon_{ijk}J_k$. So $\mathfrak{so}(1,3) \ncong \mathfrak{so}(4) \cong \mathfrak{su}(2)\oplus\mathfrak{su}(2)$ as real algebras, and because $[J_i, K_j] \neq 0$ the real algebra does not split into commuting pieces.

The Complexification Trick

Since the real algebra does not decompose, we complexify it: we allow complex linear combinations of the generators, passing to $\mathfrak{so}(1,3)_{\mathbb{C}}$, and define

$$N_i = \tfrac{1}{2}(J_i + iK_i), \qquad \tilde{N}_i = \tfrac{1}{2}(J_i - iK_i).$$

With Srednicki's boost generator $M^{i0} = -K_i$ these are his $N_i = \frac{1}{2}(J_i - iM^{i0})$ and $N_i^\dagger = \frac{1}{2}(J_i + iM^{i0})$, (33.16)–(33.17). The minus sign in $[K_i, K_j]$ is exactly what makes the two sets commute:

$$[N_i, N_j] = i\epsilon_{ijk}N_k, \qquad [\tilde{N}_i, \tilde{N}_j] = i\epsilon_{ijk}\tilde{N}_k, \qquad [N_i, \tilde{N}_j] = 0.$$

The complexified algebra therefore does decompose:

$$\mathfrak{so}(1,3)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{su}(2)_{\mathbb{C}}\oplus\mathfrak{su}(2)_{\mathbb{C}}.$$

Why this is allowed. The finite-dimensional complex representations of a real Lie algebra $\mathfrak{g}$ are in one-to-one correspondence with the complex-linear representations of its complexification $\mathfrak{g}_{\mathbb{C}}$, and irreducibility is preserved. So classifying the representations of the easy algebra $\mathfrak{su}(2)_{\mathbb{C}}\oplus\mathfrak{su}(2)_{\mathbb{C}}$, which are pairs of ordinary angular-momentum representations, classifies those of the physical algebra.

Counting dimensions. $\mathfrak{su}(2)$ is a 3-dimensional real algebra (basis σ_i). Its complexification $\mathfrak{su}(2)_{\mathbb{C}}$ is 3-dimensional over $\mathbb{C}$ and 6-dimensional over $\mathbb{R}$ (basis $\{\sigma_i, i\sigma_i\}$). So $\mathfrak{so}(1,3)_{\mathbb{C}}$ is 6-dimensional over $\mathbb{C}$, i.e. a 12-dimensional real space, with basis $\{\sigma_i^N, i\sigma_i^N\}\cup\{\sigma_i^{\tilde{N}}, i\sigma_i^{\tilde{N}}\}$.

The physical slice. The physical algebra $\mathfrak{so}(1,3)$ is a 6-dimensional real slice of this 12-dimensional space. Inverting the definitions,

$$J_i = N_i + \tilde{N}_i, \qquad K_i = -i\,(N_i - \tilde{N}_i),$$

with *real* rotation angles and rapidities multiplying them. In a finite-dimensional representation with N_i and $\tilde{N}_i$ Hermitian:

- the rotation generators J_i are Hermitian, so rotations $\exp(i\theta_j J_j)$ are unitary;
- the boost generators K_i are anti-Hermitian, so boosts $\exp(i\eta_j K_j) = \exp\big(\eta_j(N_j - \tilde{N}_j)\big)$ are not unitary. This is not a choice: a finite-dimensional representation of the non-compact Lorentz group cannot be unitary (§10.3).

The six physical generators are this specific combination, not "two sets of Pauli matrices".

A note on notation. Srednicki writes our $\tilde{N}_i$ as $N_i^\dagger$. On the Hilbert space of states, where the representation is unitary and J_i, K_i are Hermitian, this is a true adjoint. In the finite-dimensional representations below, both N_i and $\tilde{N}_i$ are represented by Hermitian matrices and $\tilde{N}_i \neq N_i^\dagger$; there the dagger is only a label. What survives is a relation between the group elements: the generators of the $(1,2)$ representation are the Hermitian conjugates of those of the $(2,1)$ ($J_i \to J_i^\dagger$, $K_i \to K_i^\dagger$), so the matrices are related by $R = (L^\dagger)^{-1}$. This R acts on the upper-index field $\psi^{\dagger\dot{a}}$; the lower-index field $\psi^\dagger_{\dot{a}}$ transforms with $L^* = \varepsilon^{-1} R\, \varepsilon$ ($\varepsilon = i\sigma_2$, the invariant symbol that raises the index). Srednicki's R, (34.12), acts on $\psi^\dagger_{\dot{a}}$; by his (34.17), $S_R = -S_L^*$, it is this L^*. LecNote16 p. 9 uses his R. Both describe the same representation; the table in §10.26 lists the matrix for each index position. Our K_i follows §10.3, $K_i = M^{0i}$, the negative of Srednicki's $K_i = M^{i0}$.

Classifying the Representations

Irreducible representations are labeled by a pair $(2n+1,\, 2n'+1)$, the dimensions under the N and $\tilde{N}$ algebras, with $n, n' = 0, \frac{1}{2}, 1, \ldots$. Since $J_i = N_i + \tilde{N}_i$, the rotation content is spins $|n - n'|, \ldots, n + n'$. Representations with $n + n'$ an integer are representations of $SO^+(1,3)$ itself; those with $n + n'$ half-integer are representations of $SL(2,\mathbb{C})$ only, and change sign under a 2π rotation.

$(2,1)$***: the left-handed Weyl spinor.*** $J_i = \frac{\sigma_i}{2}$, $K_i = -\frac{i\sigma_i}{2}$. Then $N_i = \frac{\sigma_i}{2}$ (dimension 2) and $\tilde{N}_i = 0$ (dimension 1). A left-handed spinor transforms as $\psi_L \to L\psi_L$, with $L = \exp\big(\frac{i}{2}\theta_i\sigma_i + \frac{1}{2}\eta_i\sigma_i\big) \in SL(2,\mathbb{C})$.

$(1,2)$***: the right-handed Weyl spinor.*** $J_i = \frac{\sigma_i}{2}$, $K_i = +\frac{i\sigma_i}{2}$. Then $N_i = 0$ and $\tilde{N}_i = \frac{\sigma_i}{2}$. A right-handed spinor transforms as $\psi_R \to R\psi_R$, with $R = \exp\big(\frac{i}{2}\theta_i\sigma_i - \frac{1}{2}\eta_i\sigma_i\big) = (L^\dagger)^{-1}$.

$(2,1)\oplus(1,2)$***: the Dirac spinor.*** In the Weyl basis $\Psi=(\psi_L,\psi_R)^T$ transforms with $\mathrm{diag}(L,R)$, generated by the blocks $J_i=\frac{1}{2}\mathrm{diag}(\sigma_i,\sigma_i)$, $K_i=\frac{i}{2}\mathrm{diag}(-\sigma_i,\sigma_i)$ of §10.3.

$(2,2)$***: the four-vector.*** A four-vector carries one index of each type. Following Srednicki's $\sigma^\mu=(I,\boldsymbol{\sigma})$ contracted with the lower-index x_μ, map it to a 2×2 Hermitian matrix:

$$X=x_\mu\sigma^\mu=\begin{pmatrix}-x^0+x^3 & x^1-ix^2\\ x^1+ix^2 & -x^0-x^3\end{pmatrix},\qquad \det X=(x^0)^2-|\mathbf{x}|^2=-x_\mu x^\mu.$$

It transforms as

$$X\to X'=LXL^\dagger,$$

the $(2,1)$ rule acting on the left index and its complex conjugate (equivalent to the $(1,2)$) on the right. Because $\det L=1$, $\det X'=\det X$, and X' is again Hermitian, so this is a Lorentz transformation $x\to\Lambda(L)x$ of the components; L and $-L$ give the same Λ. Its rotation content is $\frac{1}{2}\otimes\frac{1}{2}=0\oplus 1$: the time component and a spatial vector.

Representation	$(d_N,d_{\tilde{N}})$	(n,n')	**Transformation**
Left-handed Weyl	$(2,1)$	$(\frac{1}{2},0)$	$\psi_L\to L\psi_L$
Right-handed Weyl	$(1,2)$	$(0,\frac{1}{2})$	$\psi_R\to(L^\dagger)^{-1}\psi_R$
Four-vector	$(2,2)$	$(\frac{1}{2},\frac{1}{2})$	$X\to LXL^\dagger$, $X=x_\mu\sigma^\mu$
Dirac spinor	$(2,1)\oplus(1,2)$	$(\frac{1}{2},0)\oplus(0,\frac{1}{2})$	$\Psi\to\mathrm{diag}\big(L,(L^\dagger)^{-1}\big)\Psi$

THE BASIC FINITE-DIMENSIONAL REPRESENTATIONS OF THE LORENTZ GROUP, WITH $L\in SL(2,\mathbb{C})$

KEY DEFINITIONS

- **Isomorphic (≅)**: the same structure, e.g. the same commutation relations in a suitable basis.
- **Universal double cover**, e.g. $SL(2,\mathbb{C})\to SO^+(1,3)$:
  - *Cover*: a continuous, locally one-to-one projection from the covering group onto the base group that respects multiplication.
  - *Double*: the map is exactly 2-to-1.
  - *Universal*: the covering group is simply connected (every loop can be shrunk to a point). $SL(2,\mathbb{C})$ is; $SO^+(1,3)$ is not, since a 2π rotation is a loop that cannot be contracted, while a 4π rotation can.